

Week 4

RISK MANAGEMENT IN FINANCIAL INSTITUTIONS

Key objectives of this lecture

What types of risks do banks face?

How to bank manage these risks?

2

Key objectives of this lecture

“Is banking a particularly risky business? If so, what types of risks do banks face?”

In a market system, businesses of all types face risks, and many fail.

Economists and policymakers are particularly concerned about the risk and potential for failure that banks face because they play a vital role in the financial system.

In this lecture, we discuss risks facing financial institutions –entails several types of risks: credit risk, solvency risk, liquidity risk, interest rate risk, operating and reputation risk

3

Risk Management

- The ability to properly and effectively identify, assess, monitor, and manage each type of risk is critical to a bank's financial soundness and profitability.
- Diversification of business activity helps reduce the impact that volatility in any particular area may have on profits.
- Risk management policies and procedures are evolutionary and are subject to ongoing review and modification.

4

Types of Risk

- 1) Credit Risk
- 2) Solvency Risk
- 3) Liquidity Risk
- 4) Interest Rate Risk
- 5) Operating and Reputation Risk

Credit risk

- Credit risk is the possible loss that occurs when a counterparty or an issuer of securities or other instruments held by the firm fails to meet its contractual obligations.



Measuring Credit Risk

These ratios assess a bank's exposure to loan losses and how well it manages credit risk.

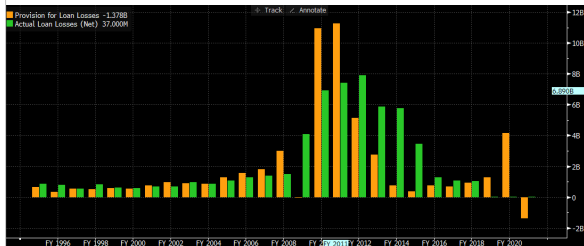
Loan Loss Reserves / Gross Customer Loans & Advances (%)

- Measures how much of the loan book is reserved for potential loan losses.
- Higher value = Lower risk** (indicates better provisioning for bad loans).

Loans & Advances to Customers / Total Assets (%)

- Measures the proportion of assets tied to loans, indicating loan concentration risk.
- Higher value = Higher risk** (more exposure to credit risk).

Bloomberg data on Lloyds bank



Bloomberg data on Lloyds bank



Managing credit risk

- The credit risk of an **individual loan** concerns the losses the bank will experience if the borrower does not repay the loan.
- The credit risk of a bank's loan portfolio concerns the **aggregate credit risk of all the loans** in the bank's portfolio.
- Banks must manage both dimensions effectively to be successful.

Managing credit risk

- **Adverse selection** in loan market: those likely to default are more likely to take loan (borrowers with risky project gain)
- **Moral hazard** in loan market: borrowers may have incentive to engage undesirable activities (likely to default)
- Bank manage credit risk by:
 - 1) Screening and monitoring
 - Screening: screen bad credit risk from good one by collecting **reliable information**.
 - Specialization lending: lend to local firm or **specific industry**
 - Monitoring and enforcement of **restrictive covenants**: monitor borrowers' activities and limit the borrower from engaging in risky activities.

Managing credit risk

- 2) Long-term customer relationship
If borrowers have long-term engagement with banks, this reduce information collection costs and make it easier to screen out bad credit risks.
- 3) Loan commitments – provide loan with special term for existing clients. (e.g: a guarantee that the lender will continue to provide funds up to the credit limit at a certain interest rate)
- 4) Collateral – provide security to banks in the event of default (e.g. property, car)
- 5) Credit rationing – provide borrowers with smaller loan to reduce default risk

Managing Credit Risk of Loan Portfolio

- **Internal Credit Risk Ratings** are used to
 - identify problem loans
 - determine adequacy of loan loss reserves; price loans
- **Loan Portfolio Analysis** is used to ensure that banks are well diversified
 - **Concentration ratios** measure the percentage of loans allocated to a given geographic location, loan type, or business type.

13

Solvency Risk

- **Definition** - The risk that a bank cannot meet its long-term financial obligations due to insufficient capital.
- **Key Characteristics**
 1. **Liabilities > Assets**
 1. Occurs when the value of a bank's liabilities exceeds its assets.
 2. **Potential Outcomes**
 1. Bankruptcy or operational failure.
 3. **Causes**
 - **High Debt Levels**
 - Thin capital base relative to liabilities.
 - **High-Risk Loans**
 - Significant loan defaults deplete capital.

Solvency Risk

- Bank with high capital

Assets	£ millions	Liabilities	£ millions
Reserve	10	Deposit	90
Loans	90	Capital	10

- Bank with low capital

Assets	£ millions	Liabilities	£ millions
Reserve	10	Deposit	96
Loans	90	Capital	4

Solvency Risk

- Let's say that these two banks have made a loan of £5 millions to a company that goes bankrupt.
- The loan value at both banks will fall by £5 millions.
- Both banks capital will also drop by £5 millions.

Solvency Risk

- Bank with high capital

Assets	£ millions	Liabilities	£ millions
Reserve	10	Deposit	90
Loans	85	Capital	5

- Bank with low capital

Assets	£ millions	Liabilities	£ millions
Reserve	10	Deposit	96
Loans	85	Capital	-1

Solvency Risk

- Bank with high capital will still have a positive capital value.
- But bank with low capital is not solvent (bankrupt).
- This example shows how capital becomes a cushion against the decline in the value of bank assets.
- So, to be resilient against failure, banks need a large enough buffer of capital. The riskier a bank's assets, the more capital it needs.
- However, there is trade-off between safety and returns to equity holders:
 - Benefits the owners of a bank by making their investment safe
 - Costly to owners of a bank because the higher the bank capital, the lower the return on equity

Solvency Risk

- **Process of Insolvency**

1. Loans go bad (cannot be repaid).
2. Losses erode the capital base.
3. Liabilities surpass assets.
4. Bank becomes insolvent.

- **Impact**

- Loss of stakeholder confidence.
- Potential collapse of the bank.
- Broader financial system risks.

Measuring Solvency Risk

These ratios assess how well-capitalized a bank is to absorb financial shocks.

- **Total Equity / Total Assets (%)**

- Indicates how much of the bank's assets are funded by shareholders rather than debt.
- **Higher value = Lower risk** (stronger equity position).

- **Total Capital Adequacy Ratio (%)**

- Measures the bank's capital buffer against risk-weighted assets.
- **Higher value = Lower risk** (stronger capital position).

Liquidity Risk

- **Liquidity problems arise due to interactions between funding and the asset side of the balance sheet — when a bank does not hold sufficient cash (or assets that can easily be converted into cash) to repay depositors and other creditors.**
- **The demands for liquidity include accommodating deposit withdrawals and paying other liabilities as they come due.**

Liquidity Risk

Assets		Liabilities	
Reserve	xxx	Unstable funding (e.g. short term wholesale funding)	xxx
Cash	xxx	Stable funding (e.g. long term bonds)	xxx
Securities	xxx	Capital	xxx
Loans	xxx		

22

- A large amount of short-term, unstable funding relative to liquid assets such as cash and securities means that if all investors attempt to withdraw this short-term funding at the same time, then the bank's buffer of liquid assets is quickly depleted.
- Since the bank's other, less liquid assets (such as loans) are not due to be repaid soon, the bank cannot give all of the investors their money.
- **Fire sale** (selling all assets at a great discount) can result in bank default.

Liquidity Risk

To avoid liquidity problems banks need to have a combination of:

- Stable sources of funding that will not dry up during stressed market conditions.
- A buffer of liquid assets

23

Measuring Liquidity Risk

These ratios measure how well a bank can meet its short-term obligations.

• Loans & Advances to Customers / Deposits & Short-Term Funding (%)

- Measures reliance on deposits for lending.
- **Higher value = Higher risk** (suggests potential liquidity issues if too many loans are issued relative to deposits).

• Liquid Assets / Deposits & Short-Term Funding (%)

- Measures the proportion of deposits backed by liquid assets.
- **Higher value = Lower risk** (stronger liquidity position).

Interest Rate Risk

- It is the riskiness of earnings and returns that is associated with changes in interest rates.
- If a bank has liabilities that are more sensitive to interest rate changes than assets:
 - an increase in interest rates will reduce bank profits.
 - a decrease in interest rates will raise bank profits.
- Basic gap analysis:

$$= (\text{rate sensitive assets} - \text{rate sensitive liabilities}) \times \Delta \text{ interest rates}$$

$$= \Delta \text{ in bank profit}$$

Interest Rate Risk

Assets		Liabilities	
Rate sensitive assets (e.g. variable rate and short-term loans; short-term securities)	£20 million	Rate sensitive liabilities (e.g. variable rate certificate of deposit (CDs) ; money market deposit)	£50 million
Fixed rate assets (e.g. reserves; long-term loans; long-term securities)	£80 million	Fixed-rate liabilities (e.g. checkable deposit, savings deposit, long term CDs)	£40 million
		Capital	£10 million

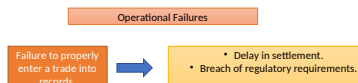
Example for Bank A

- Has £20 million assets that are rate-sensitive (with interest rates that change at least once a year), and £80 million assets are that have fixed-rate (with interest rates that remain unchanged for over a year).
- Has £50 million of rate-sensitive liabilities and £40 million of fixed-rate liabilities.
- Suppose that interest rates rise by 5% on average, from 10% to 15%.
 - The income on the assets rises by £1 million (5% x £20 million of rate-sensitive assets), while the payments on the liabilities rise by £2.5 million (5% x £50 million of rate-sensitive liabilities).
 - The Bank's profits now reduce by £1.5 million (£1 million - £2.5 million).
 - However, if interest rates fall by 5%, we will find that the Bank's profits increase by £1.5 million.

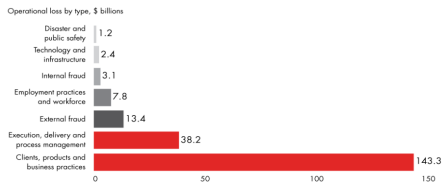
26

Operating and Reputation Risk

- The risk of direct or indirect loss resulting from inadequate or failed internal processes, people, and systems, or from external events.
- Includes losses from fraud, theft, terrorism, litigation, and even reputation problems.
- An operational failure may result in financial loss, regulatory risk, or damage to a firm's reputation.

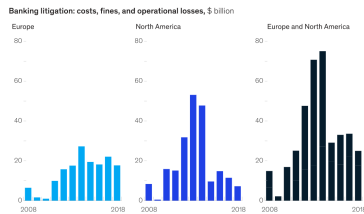


From 2011 to 2016, major banks suffered nearly \$210 billion in losses from operational risk



<https://www.bain.com/insights/how-banks-can-manage-operational-risk/>

Operational Risk has increased since the global financial crisis and has remained high.



Source: McKinsey & Company

<https://www.mckinsey.com/capabilities/risk-and-resilience/our-insights/the-future-of-operational-risk-management-in-financial-services>

INTERPOL

Who we are Crimes How we work Our partners What you can do News Wanted persons

INTERPOL report shows alarming rate of cyberattacks during COVID-19

4 August 2020

COVID-19 IMPACT

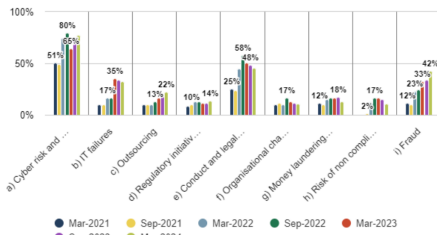
Home > News and Events > News > 2020 > INTERPOL report shows alarming rate of cyberattacks during COVID-19

Shift in targets from individuals to governments and critical health infrastructure

According to the industry body UK Finance, British banks paid out £147m in 2020 reimbursing losses caused by “authorised” scams, where customers unwittingly make payments from their accounts to criminals. But roughly double that amount was lost overall to this type of fraud.

<https://www.ft.com/content/30553ae9-cdfd-483c-a1ef-c04e3135f9da>

Figure 49: Main drivers of operational risk as seen by banks *



Source: [Risk assessment report July 2024.pdf](#)

Managing Operational Risk

- The financial crisis precipitated a wave of regulatory fines and enforcement actions on misselling, questionable mortgage-foreclosure practices, financial crimes, London Inter-bank Offered Rate (LIBOR) fixing, and foreign-exchange misconduct.
- Compared to other risks, operational risks are more complex as it involves diverse risk types that involves different groups of people (compliance, financial crime, cyber risk, and IT risk).
- Operational risk can be mitigated through corporate governance.
- Operational-risk management requires oversight and transparency of almost all organizational processes and business activities.
- This means that as long as people, systems, and processes remain imperfect, operational risk cannot be fully eliminated.
- The Basel Committee on Banking Supervision (BCBS) has highlighted the importance of operational risk.
- They can be mitigated by good corporate governance – putting in place foundational elements such as loss-event reporting and risk-control self-assessments and holding capital buffer against operational risk.

32

Dilemma: Profitability versus Safety

- A bank must balance the demands of three constituencies:
 - shareholders**—mainly interested in profitability
 - depositors**—mainly interested in safety
 - regulators**—mainly interested in safety
- One way for a bank to increase expected profits is to take on more risk. However, this can jeopardize bank safety.

Solvency and Liquidity

- **Solvency:** Maintaining the momentum of a going concern, attracting customers and financing.
 - A firm is insolvent when the value of its liabilities exceeds the value of its assets.
- **Liquidity:** The ability to fund deposit withdrawals, loan requests, and other promised disbursements when due.
 - A bank can be profitable and still fail because of illiquidity.

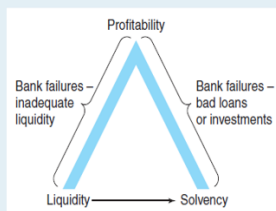
34

Conflicting Demands

- A bank must balance profitability, liquidity, and solvency.
- Bank failure can result from excessive losses on loans or securities → from over-aggressive profit seeking. But a bank that only invests in high-quality assets may not be profitable.
- Failure can also occur if a bank cannot meet liquidity demands. If assets are profitable but illiquid, the bank also has a problem.
- Bank insolvency often leads to bank illiquidity.

35

Profitability Goal Versus Liquidity and Solvency



The goal of bank management is to maximize the value of the firm. However, higher profits must not be achieved at the expense of bank safety

36

How a Capital Crunch Caused a Credit Crunch During the Global Financial Crisis

- Shortfalls of bank capital led to slower credit growth
 - Huge losses for banks from their holdings of securities backed by residential mortgages.
 - Losses reduced bank capital
- Banks could not raise much capital on a weak economy, and had to tighten their lending standards and reduce lending.

References

- Mishkin, F.S., 2018. Financial Market and Institutions, Global Edition, Chapter 23, Risk Management in Financial Institutions. (available online at UWE library)
- Lundtofte, F., & Nielsen, C. Y. (2019). The Effect of Stricter Capital Regulation on Banks' Risk Taking: Theory and Evidence. European Financial Management, 25(5), 1229-1248. <https://doi.org/10.1111/eufm.12205>
